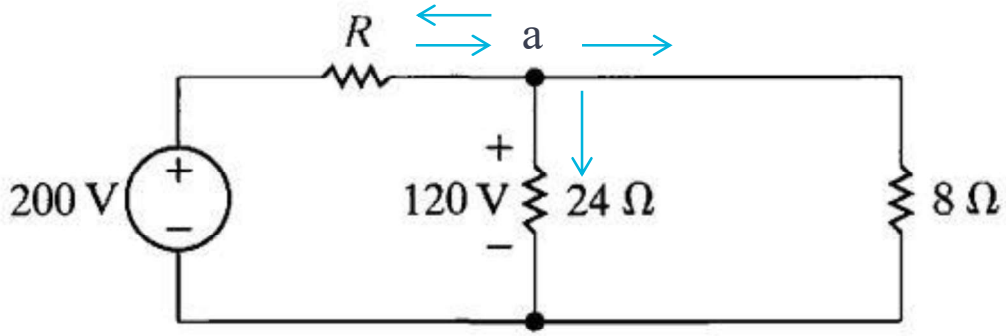
General Practices

- Example-1: Find the value of R in the circuit.



- Writing a KCL equation for the upper essential node:

$$-\left(\frac{200-120}{R}\right)+\frac{120}{24}+\frac{120}{8}=0$$

$$-\left(\frac{200-120}{R}\right)+5+15=0$$

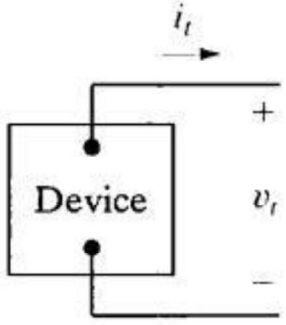
$$\frac{80}{R}=20$$

$$\frac{80}{20}=R$$

$$R=4\,\Omega$$

General Practices

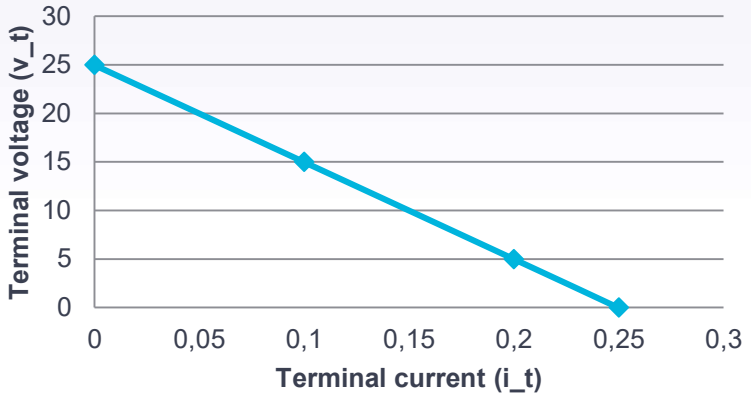
- Example-2: The terminal voltage an terminal current were measured on the device shown. Usign these values, create a straight line plot of v_t versus i_t. Compute the equation of the line and use the equation to construct a circuit model for the device.



(a)

v_t (V)	i_t (A)
25	0
15	0.1
5	0.2
0	0.25

(b)



The equation of a straight (linear) line with gradient m and intercept c on the y-axis is

$$y = mx + c$$

It is easy to observe that: $c = 25$

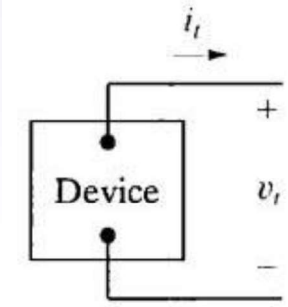
Calculation of m:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{25-15}{0-0.1} = -100$$

$$y = -100x + 25$$

General Practices

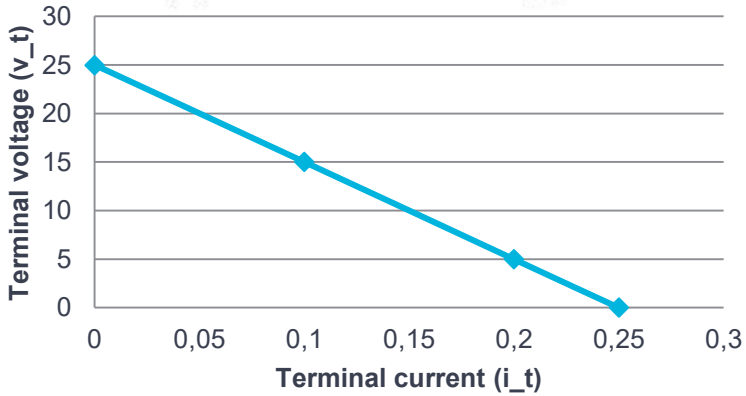
- Example-2: The terminal voltage an terminal curret were measured on the device shown. Using these values, create a straight line plot of v_t versus i_t. Compute the equation of the line and use the equation to construct a circuit model for the device.



(a)

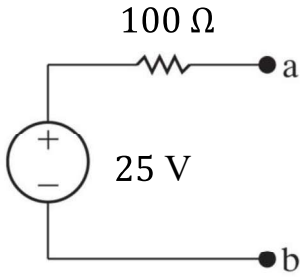
v_t (V)	i_t (A)
25	0
15	0.1
5	0.2
0	0.25

(b)



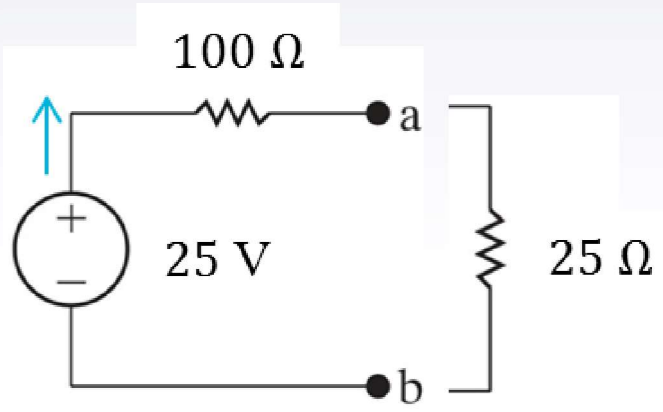
$$y = -100x + 25$$

- The device is able to supply 25 V even when there is no current (voltage source behavior).
- As the current drawn from the device increases, the terminal voltage linearly drops by 100 times of the drawn current (resistor behavior).



General Practices

- Example-2: b) Use the model constructed in (a) to predict the power that the device will deliver to a 25 Ω resistor.



$$i = \frac{25}{125} = 0.2\ A$$

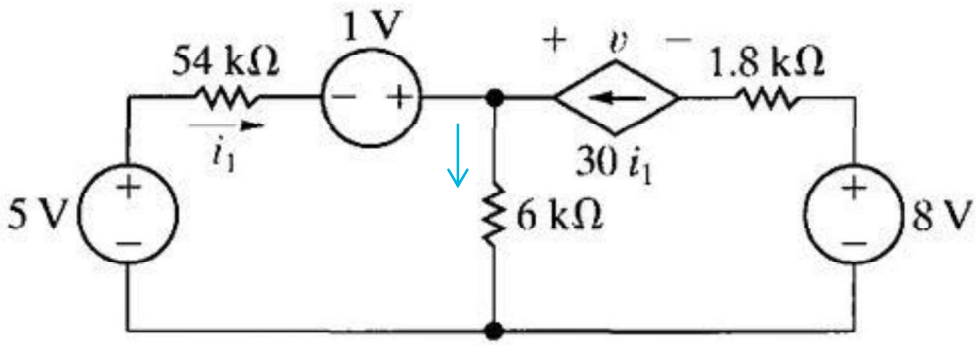
$$p_{25} = i^2(R) = 0.04(25) = 1\ W$$

$$p_{100} = i^2(R) = 0.04(100) = 4\ \text{W}$$

$$p_{25V} = -i(v) = -(0.2)25 = -5\text{W}$$

General Practices

- Example-3: For the given circuit, find (a) the current i_1, (b) the voltage v, (c) the total power generated, and (d) the total power absorbed.



a) Writing a KCL for the upper essential node

$$-i_1 + i_{6k\Omega} - 30i_1 = 0$$

$$i_{6k\Omega} = 31i_1$$

Writing a KVL for loop at the left, following a clockwise direction

$$-5 + 54000\, i_1 - 1 + (6000)31i_1 = 0$$

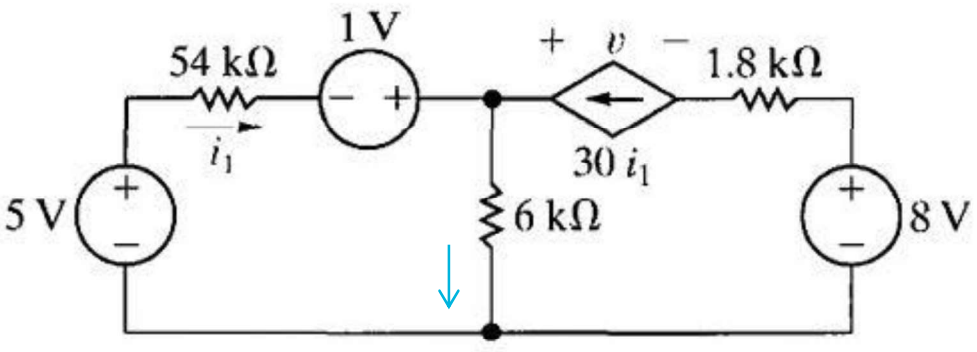
$$54000\, i_1 + 186000i_1 = 6$$

$$240000i_1 = 6$$

$$i_1 = \frac{6}{240000} = 0.000025A = 25\ \mu A$$

General Practices

- Example-3: For the given circuit, find (a) the current i_1, (b) the voltage v, (c) the total power generated, and (d) the total power absorbed.



b) Writing a KVL for the loop at the right, following a clockwise direction

$$v - (1800)30i_1 + 8 - (6000)31i_1 = 0$$
$$v - 54000i_1 + 8 - 186000i_1 = 0$$
$$-v = 8 - (240000)i_1$$
$$v = -8 + (240000)(25)(10^{-6})$$
$$v = -8 + 6$$
$$v = -2\ V$$

c) $p_{5V} = -(5)(25)(10^{-6}) = -125(10^{-6})\ W$

$p_{5V} = -125\ \mu W$

$p_{54k\Omega} = (54000)[(25)(10^{-6})]^2 = 33.75\ \mu W$

$p_{1V} = -(1)(25)(10^{-6}) = -25\ \mu W$

$p_{6k\Omega} = (6000)[(31)(25)(10^{-6})]^2$

$= 3603.75\ \mu W$

$p_{30i_1} = -(-2)(30)(25)(10^{-6}) = 1500\ \mu W$

$p_{1.8k\Omega} = (1800)[(30)(25)(10^{-6})]^2$

$= 1012.5\ \mu W$

$p_{8V} = -(8)(30)(25)(10^{-6}) = -6000\ \mu W$